

Randomized and low-rank approximation

Spectral-gap-independent constants for randomized block polynomial interpolation

Difficulty: challenging **Importance:** interesting to the community

Status: [Lean verified](#)

Rating rationale: Challenging because almost-sure invertibility does not yield spectrum-uniform probabilistic constants; community impact is explaining cluster robustness of block Lanczos.

Negative resolution - 12 September 2026 (UTC)

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The answer to the displayed uniform-probability question is **no**. The [exact target](#) and [Sections 1-5 of the complete proof](#) give deterministic admissible blocks with $b = 2, d = 3$,

$$\Lambda_1 = 2I_2, \quad \Lambda_2 = \text{diag}(\varepsilon, 2\varepsilon), \quad \Lambda_3 = \text{diag}(0, 1).$$

Along $\varepsilon_m = 1/(m + 5)$, the probability that $\chi_{\text{mono}}\chi_{\text{coef}}$ is below any fixed finite constant tends to zero. This contradicts the requested bound already at $\delta = 1/2$. The proof uses the literal recurrence and all prescribed root orderings; no input depends on the sampled Gaussian matrices.

[Proof PDF](#) · [Standalone proof TeX](#) · [Independent mathematical review](#) · [Submission and public-source audit](#).

The complete proof passed a separate Codex-agent informal audit in September 2026. Substantial AI assistance is disclosed. That informal audit did not provide formal verification; the later Lean evidence is recorded separately below. External human peer review is not claimed. Shao retains credit for the original conjecture and interpolation framework. The counterexample uses interlaced block spectra and a repeated eigenvalue within one block, both permitted by the original hypotheses. It does not address a different question imposing ordered spectral intervals or refute cluster-robust block Lanczos convergence itself. The original statement, permanent ID, references and dated history remain below; the ratings are historical.

Lean proof and verification evidence

Formalization: Sidney Holden, with OpenAI Codex assistance, under Apache-2.0. **Mathematical proof and integration/verification submission:** George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology. Nian Shao retains credit for the original framework and conjecture. The [complete proof at immutable revision 414371c9](#) reuses Holden’s implementation without changing its 23 mathematical Lean files.

The formal theorem negates the full original uniformly bounded-in-probability statement. For every finite real constant C , it constructs deterministic admissible data with $b = 2, d = 3$ such that

$$\Pr\{\chi_{\text{mono}}\chi_{\text{coef}} \leq C\} < \frac{1}{2}.$$

The literal descending recurrence, every prescribed root ordering, independent standard real Gaussian entries, actual Euclidean operator norms, endpoint 0/0 convention and spectrum-independent quantifiers are retained. Almost-sure validity is proved for all admissible data; totalizing invalid samples does not alter any probability. The formal sequence $\varepsilon_m = 1/(m + 6)$ for $m \geq 0$ is the manuscript sequence with its index shifted by one. Exact two-by-two algebra, a polynomial null-set argument and measure convergence avoid interval searches; the rational witness alone is not used as a positive-probability event.

All **ten declarations** are listed in [formalization.yaml](#). The final results are:

- *NLA.KE05.no_finite_uniform_constant*: each finite bound fails for some deterministic admissible input.
- *NLA.KE05.not_uniform_probability_conjecture*: the complete original conjecture is false.

The literal recurrence, validity, exact witness, lower bound, almost-sure divergence and marginal probability theorems establish these conclusions. The [independent Challenge](#), [definitions](#), [frozen numerical targets](#) and [proof correspondence](#) record their meaning. The historical statements-only wording in the frozen targets is retained as evidence; current status is stated here.

The project pins **Lean 4.33.1**, Mathlib [0df444a360eaa60ab8c11dca51a86af692955474](#) and LeanCert [621a43d7cf21f87872392a01e874f2f1dbddc926](#). On **16 September 2026 UTC**, [Linux run 35058392398](#), [verification job 104673346252](#) checked literal proof commit [414371c9a76aafd7477d9705f9644f7efeb5e329](#). All ten LeanCert kernel-trust assertions, Comparator statement checks and Lean default-kernel replay passed. Every exported theorem’s transitive axiom report contains only *propext*, *Classical.choice* and *Quot.sound*. Required sandbox, rejection and checker controls ran within the successful project job. [Actual logs and source hashes](#) bind all **151** candidate inputs.

Two independent AI-agent [complete-source reviews](#) and an [audit of the fresh actual execution](#) support this record. The checks ran on GitHub’s non-root Linux runner; no local macOS rerun or external human peer review is claimed. See the [project README](#) for *lake build Solution* and the shared Linux reproduction commands. Later publication and merge revisions require their own checks; the named proof run does not certify a later commit.

Original statement (retained)

Fix integers $b \geq 1, d \geq 2$. Let $\Lambda_1, \dots, \Lambda_d \in \mathbb{R}^{b \times b}$ be diagonal with pairwise disjoint spectra. Draw all entries of $\Omega_1, \dots, \Omega_d \in \mathbb{R}^{b \times b}$ independently from $N(0, 1)$ and put $B_i = \Omega_i^{-1} \Lambda_i \Omega_i$. Let a and c be the smallest and largest diagonal entries in the entire family.

For each $k \in \{1, \dots, d\}$, reorder the triples $(B_i, \Lambda_i, \Omega_i)$ into the order $(k, 1, \dots, k-1, k+1, \dots, d)$ and use superscript (k) for that ordering. In descending order $i = d, d-1, \dots, 1$, compute

$$S_{i,i}^{(k)} = I_b, \quad S_{i,j}^{(k)} = B_i^{(k)} S_{i,j-1}^{(k)} - S_{i,j-1}^{(k)} \hat{B}_j^{(k)} \quad (j = i+1, \dots, d),$$

$$\hat{\Omega}_i^{(k)} = \Omega_i^{(k)} S_{i,d}^{(k)}, \quad \hat{B}_i^{(k)} = (\hat{\Omega}_i^{(k)})^{-1} \Lambda_i^{(k)} \hat{\Omega}_i^{(k)}.$$

All norms below are spectral norms. Define

$$\chi_{\text{mono}}^{(k)} = \max_{2 \leq i \leq d} \left\{ 1, \frac{\|aI - \hat{B}_i^{(k)}\|_2}{\|aI - \Lambda_i^{(k)}\|_2}, \frac{\|cI - \hat{B}_i^{(k)}\|_2}{\|cI - \Lambda_i^{(k)}\|_2} \right\},$$

$$\chi_{\text{coef}}^{(k)} = \|(S_{1,d}^{(k)})^{-1}\|_2^{1/(d-1)} \min_{\substack{2 \leq i \leq d, \lambda \in \sigma(\Lambda_1^{(k)}) \\ \eta \in \sigma(\Lambda_i^{(k)})}} |\lambda - \eta|, \quad \chi_{\text{mono}} = \max_k \chi_{\text{mono}}^{(k)}, \quad \chi_{\text{coef}} = \max_k \chi_{\text{coef}}^{(k)}.$$

The inverses exist almost surely, as shown in the source. If an endpoint ratio is $0/0$, set it to 1; its matrix is necessarily a scalar matrix and the surrounding maximum already includes 1.

Is the family of random variables $\chi_{\text{mono}} \chi_{\text{coef}}$ uniformly bounded in probability over all admissible diagonal data? Precisely, for each $0 < \delta < 1$, does there exist a finite $C(b, d, \delta)$ such that, for every fixed admissible $(\Lambda_1, \dots, \Lambda_d)$,

$$\Pr\{\chi_{\text{mono}} \chi_{\text{coef}} \leq C(b, d, \delta)\} \geq 1 - \delta?$$

The constant must be independent of the eigenvalues and all gaps within and between their blocks. Probability is taken separately for each fixed input, not for one draw required to work simultaneously for every spectrum. This quantifies the source's "with high probability" statement with its explicitly allowed failure-probability dependence.

These constants control matrix-polynomial interpolation in the convergence analysis of randomized small-block Lanczos. The bound would explain robustness to clusters much larger than the starting block. The scalar-block case is already bounded deterministically; noncommutativity is the unresolved obstacle.

References

N. Shao, *A structural bound for cluster robustness of randomized small-block Lanczos*, arXiv:2507.10144v2, 30 May 2026, Theorem 1, equation (12), and Conjecture 1 in §3.2 ([primary full text](#)). T. Chen et al., *Does block size matter in randomized block Krylov low-rank approximation?*, arXiv:2508.06486, §§4,6 ([primary paper](#)), is related: its conjecture changes spectral-gap dependence in output bounds, whereas this question bounds the intermediate random interpolation constants themselves.

Status check — 2026-09-08

Latest arXiv abstract/history and May 2026 v2 inspected. The revision explicitly retains Conjecture 1, including independence of gaps within and between blocks. Searches for author/title with "proof", "conjecture", "cluster robustness", and 2026 found no resolution. Generic nonsingularity of the recurrence does not give the claimed uniform probabilistic bound.

Audit — 2026-09-10

Rechecked [Shao's May 2026 Conjecture 1](#) and current version history. It still requires independence of gaps inside and between blocks. Cluster-robustness and later-author searches found no resolution. The scalar-block baseline does not resolve noncommuting block interpolation.

Resolution audit - 2026-09-12

The independent review checked the full algebra, almost-sure nonsingularity of every prescribed ordering, and the deterministic-input probability quantifiers. A fresh public fork/branch/PR audit and primary-source search found no prior resolution of KE-05 at the recorded time; its exact scope and limits are in the linked submission record. All original mathematical hypotheses remain unchanged.